

UNIMAG Interface – Power Supply EPICS IOC Specification

Process Variables (PVs)

Current Control

- **Current Readback (`CURRENT_RB`)**
Provides the measured output current of the magnet channel.
PV format:
 - `$ (P) : $ (R) : CURRENT_RB`
 - **Current Setpoint (`CURRENT_SP`)**
Sets the desired current for the magnet channel.
PV format:
 - `$ (P) : $ (R) : CURRENT_SP`
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Channel State Control

- **Channel State Readback (`STATE_RB`)**
Reports the current operational state of the magnet channel.
PV format:
 - `$ (P) : $ (R) : STATE_RB`
 - **Channel State Setpoint (`STATE_SP`)**
Allows the user to request a change of the channel state (e.g. via ON, STANDBY, RESET commands).
PV format:
 - `$ (P) : $ (R) : STATE_SP`
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Configuration Parameters

- **set_tolerance**
Allowed tolerance on `CURRENT_RB` when checking if the requested current has been reached.
Default: 1
 - **zero_tolerance**
Allowed tolerance when checking that the current is zero.
Default: 1
 - **set_timeout_s**
Timeout (in seconds) to verify that the requested current has been reached.
If the timeout expires, the operation is considered failed.
Default: 10
 - **max_retries**
Number of retries before declaring `STATE_NOT_REACHED`.
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Channel State Values (`STATE_RB`, `STATE_SP`)

The channel state PVs represent the operational status of the magnet channel.

Possible values:

- `OFF` – Magnet is powered off
 - `ON` – Magnet is powered and active
 - `STANDBY` – Magnet is in standby mode
 - `FAULT` – A fault condition has been detected
 - `RESET` – Reset operation in progress
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UNIMAG Machine State

- **UNIMAG State Readback (`UNIMAG_STATE_RB`)**
Reports the internal UNIMAG state machine status.
PV format:
 - `$ (P) : $ (R) : UNIMAG_STATE_RB`

Possible Values

- `OK`
The power supply is ready to accept new commands.
 - `STATE_NOT_REACHED`
The requested state or current was not reached within tolerance and timeout.
This is a *low-level warning*; the machine remains ready to accept new commands.
 - `ZERO_STBY`
The magnet is ramping down to zero current and transitioning to standby.
 - `CHANGE_POL`
A polarity change is in progress.
 - `GOING_TO_SET`
The magnet is ramping toward the requested current setpoint.
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UNIMAG State Machine Behavior

Current Setpoint Handling (`CURRENT_SP`)

When a new value is written to `CURRENT_SP`, the UNIMAG state machine operates as follows:

1. **Polarity Check**
 - The controller checks whether a polarity change is required.
 - If a polarity change is needed, the state transitions to:
 - `ZERO_STBY` → `CHANGE_POL`

2. Ramp to Zero (if required)

- The current is ramped to zero.
- Zero is considered reached when:
 - $|CURRENT_RB| \leq zero_tolerance$
- The ramp must complete within `set_timeout_s`.
- If zero is not reached within the timeout (after `max_retries` attempts), the state becomes:
 - `STATE_NOT_REACHED`

3. Set Current

- If the current polarity already matches the desired polarity:
 - The system transitions directly to:
 - `GOING_TO_SET`
 - The current is ramped to the requested value.
 - Success condition:
 - $|CURRENT_RB - CURRENT_SP| \leq set_tolerance$
 - If `set_tolerance` is 0, no verification is performed and the state transitions directly to `OK`.

4. Polarity Mismatch

- If the current polarity does not match the requested polarity:
 - Ramp to zero (`ZERO_STBY`)
 - Perform polarity change (`CHANGE_POL`)
 - Then proceed with step 3.

Transition from ON to STANDBY

- If transitioning from ON to STANDBY:
 - If:
 - $|CURRENT_RB| > zero_tolerance$

the system must first ramp to zero (`ZERO_STBY`).
 - If the current is already within zero tolerance, the system can transition directly to STANDBY.

If you want, I can also:

- Convert this into **formal EPICS documentation**
- Define **state diagrams**
- Suggest **PV naming refinements**
- Align it with **EPICS record types** (bo, ao, mbbi, calc, etc.)

Just tell me 👍

can you add the retry logic on state transitions `max_retries`, can you suggest refinements?

Below is an **extended and refined version** of the specification that **explicitly defines the retry logic** (`max_retries`) for state transitions, clarifies failure semantics, and suggests **architectural refinements** that fit well with EPICS IOC design and operations practice.

UNIMAG Interface – Power Supply EPICS IOC

State Machine, Retry Logic, and Refinements

Retry Logic (`max_retries`)

`max_retries` defines how many times a state transition or verification step is retried before declaring a failure.

General Rules

- Each *verifiable transition* (e.g. ramp-to-zero, ramp-to-set) is attempted up to:
 - `max_retries + 1`
- total times (initial attempt + retries).
- A retry is triggered when:
 - The expected condition is not met within `set_timeout_s`
 - Or the power supply reports an incompatible or invalid state
- Between retries:
 - The system returns to a **safe intermediate state** (`ZERO_STBY` or `STANDBY`)
 - No user command is required to trigger retries
- If all retries fail:
 - `UNIMAG_STATE_RB` is set to `STATE_NOT_REACHED`
 - `STATE_RB` remains unchanged
 - The IOC remains responsive to new commands

Retry-Enabled State Transitions

1. Ramp to Zero (`ZERO_STBY`)

Entry Condition

- Polarity change required
- Transition from `ON` → `STANDBY` with `|CURRENT_RB| > zero_tolerance`

Success Condition

$|CURRENT_RB| \leq zero_tolerance$

within `set_timeout_s`

Retry Behavior

- If timeout expires:
 - Increment retry counter
 - Re-issue zero-current command
 - Remain in `ZERO_STBY`
 - After `max_retries` failures:
 - Transition to `STATE_NOT_REACHED`
 - Exit retry loop
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2. Polarity Change (`CHANGE_POL`)

Entry Condition

- Successful completion of `ZERO_STBY`
- Requested polarity differs from current polarity

Success Condition

- Hardware reports correct polarity

Retry Behavior

- If polarity does not change within `set_timeout_s`:
 - Retry polarity change
 - After `max_retries` failures:
 - Transition to `STATE_NOT_REACHED`
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3. Ramp to Setpoint (`GOING_TO_SET`)

Entry Condition

- Correct polarity is established

Success Condition

$|CURRENT_RB - CURRENT_SP| \leq set_tolerance$

within `set_timeout_s`

(or immediate success if `set_tolerance == 0`)

Retry Behavior

- On timeout:
 - Increment retry counter
 - Re-issue setpoint command
- After `max_retries` failures:
 - Transition to `STATE_NOT_REACHED`

Retry Counter Handling

- Retry counters are:
 - **Local to each transition**
 - Reset upon successful completion of that transition
- Retry counters are also reset when:
 - A new `CURRENT_SP` is written
 - A new `STATE_SP` command is issued

Failure Semantics (`STATE_NOT_REACHED`)

When `STATE_NOT_REACHED` is entered:

- The power supply remains in the **last known safe electrical state**
- `UNIMAG_STATE_RB = STATE_NOT_REACHED`
- The IOC:
 - Accepts new commands
 - Does **not** auto-retry unless a new command is issued
- A transition to `RESET` or a new `CURRENT_SP` clears the error

Suggested Refinements

1. Explicit Intermediate State Tracking

Suggestion:

Expose the active sub-state via an internal or optional PV:

`$ (P) : $ (R) : UNIMAG_STEP_RB`

Possible values:

- `CHECK_POL`
- `RAMP_TO_ZERO`
- `POLARITY_CHANGE`
- `RAMP_TO_SET`
- `VERIFY`

This improves:

- Debugging
 - Operator understanding
 - Alarm correlation
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2. Explicit Alarm Severity Mapping

Map failures to EPICS alarm severities:

Condition	PV	Severity
Timeout (retrying)	UNIMAG_STATE_RB	MINOR
STATE_NOT_REACHED	UNIMAG_STATE_RB	MAJOR
FAULT	STATE_RB	INVALID

3. Add a Busy Indicator

Add:

`$ (P) : $ (R) : BUSY_RB`

- 1 → State machine active
- 0 → Idle / ready

Useful for:

- GUIs
 - Script synchronization
 - Preventing overlapping commands
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4. Command Arbitration Rules

Refinement:

- If a new `CURRENT_SP` arrives while `BUSY_RB = 1`:
 - Abort current sequence
 - Reset retry counters
 - Restart state machine from polarity check

This avoids deadlocks and race conditions.

5. Optional Retry Backoff

Add (optional):

`retry_delay`

- Delay (seconds) between retries
- Prevents hammering slow or overloaded power supplies

6. Formal Transition Table (Optional)

If you want to formalize further, define transitions as:

From	Trigger	To	Verify	Retry
ON	CURRENT_SP	GOING_TO_SET	$\Delta I \leq \text{err}$	yes
ON	POL mismatch	ZERO_STBY	$I \approx 0$	yes
ZERO_STBY	Zero reached	CHANGE_POL	pol ok	yes
CHANGE_POL	Pol OK	GOING_TO_SET	$\Delta I \leq \text{err}$	yes